Technical review

Item #6 · leaf-revoked.crt · Run ID: 20261113T232802Z-2f0fe5

Path	`R:\evidence\certs\leaf-revoked.crt`
Size (bytes)	1574
Last modified (UTC)	2026-11-12T06:00:00Z
Extension	.crt
Detected type (signature)	X.509 certificate
Status	Problem found
Case	RV-SAMPLE-001
Evidence ID	E0006
Case registry	Verified: identical to the registered evidence
Registered at intake (UTC)	2026-11-13T23:27:52.795Z

Hashes

sha256	`7a16fdab1650af5675aa07f5d92d0704d18da6f87d8e76d2ff5d8b2f044e8f19`
md5	`464eff643bd286cf0b215b6521e1ae7c`

Findings

[High] Certificate revoked (CERT_REVOKED)

What it shows	C=CO, O=Rakvor Lab, CN=revoked.lab.rakvor.test (serial 2004) is listed as revoked since 2026-02-01T00:00:00Z, reason keyCompromise, according to item:3.
What it does not prove	That the holder did anything wrong (revocation also follows routine changes such as a superseded key) or that signatures made before 2026-02-01T00:00:00Z are invalid.

Method

Each file was opened read-only, hashed in a single pass with the selected algorithms, and checked against a table of file signatures. Every action was written to a hash-chained audit log.

Audit trail

Audit entries	#26 – #29
Log folder	`R:\case\runs\rv-certcheck\20261113T232802Z-2f0fe5\logs`
Digital signature	This PDF is not digitally signed.

Limitations

- Every chain is judged at one instant (--at, recorded in each report); the same certificate can be valid at one date and expired or revoked at another.
- Offline, revocation is checked only against the CRLs and OCSP responses supplied (--crl, --ocsp, or found among the evidence); a revocation published elsewhere is not seen.
- Trust lists are used as dated snapshots (date and SHA-256 recorded); services added, suspended or withdrawn after the snapshot are not reflected.
- In version 0.1 the trust list's own XAdES signature is not verified; the snapshot's source and SHA-256 are recorded so it can be checked independently. The Colombian list is a curated file, because ONAC publishes no

machine-readable list.

- Certificates issued by Rakvor's internal PKI are for internal use only: they are not a firma digital under Ley 527 de 1999, art. 28 (Colombia), and not qualified certificates under eIDAS (Regulation (EU) No 910/2014).
- The tool's results have been compared with OpenSSL and Python reference implementations on test sets, but no error rate has yet been measured on a validated corpus.
- A result without findings does not prove that a file is authentic or unaltered before acquisition.
- Modification times come from the file system of the examined copy and may differ from the original source.
- A trusted timestamp of the audit log shows that the log existed in this exact form at the stated time; it does not show that the examined copy matches the original source.